Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The number of ways in which n distinct objects can be put into two different boxes so that no box remains empty, is

(A) $2^n - 1$

(B) $n^2 - 1$

(C) $2^n - 2$

(D) $n^2 - 2$

Q.2

The number of parallelograms that can be formed from a set of four parallel lines intersecting another set of three parallel lines is

(A) 6

(B) 18

(C) 12

(D) 9

Q.3

The number of ways is which an examiner can assign 30 marks to 8 questions, giving not less than 2 marks to any question is

(A) ${}^{21}C_7$

(B) ${}^{22}C_7$

(C) ${}^{21}C_8$

(D) none of these

Q.4

The number of four-digit numbers strictly greater than 4321 that can be formed from the digits $0, 1, 2, 3, 4, 5$ allowing for repetition of digits is

(A) 310

(B) 360

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(C) 288

(D) none of these

Q.5

A closet has 5 pairs of shoes. The number of ways in which 4 shoes can be chosen from it so that there will be no complete pair is

(A) 80

(B) 160

(C) 200

(D) none of these

Q.6

The number of ways of dividing 15 men and 15 women into 15 couples, each consisting of a man and a woman, is

(A) 1240

(B) 1840

(C) 1820

(D) 2005

Q.7

A class contains 4 boys and g girls. Every Sunday five students, including at least three boys go for a picnic to Appu Ghar, a different group being sent every week. During, the picnic, the class teacher gives each girl in the group a doll. If the total number of dolls distributed was 85, then value of g is

(A) 15

(B) 12

(C) 8

(D) 5

Q.8

45 identical chocolates are distributed among 3 students in such a way that every student receives at least one chocolate and no student receives the same number of chocolates. Find the total number of ways of distribution?

(A) 946

(B) 882

(C) 883

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(D) 945

Q.9

The number of different necklaces formed by using $(2n)$ identical pearls and 3 different jewels when exactly two jewels are always together is

(A) $6n$

(B) $6n-3$

(C) $6n-6$

(D) none of these

Q.10

Let $t_n = n(n!)$. Then $\sum_{n=1}^{15} t_n$ is equal to

(A) $15!-1$

(B) $15!+1$

(C) $16!-1$

(D) none of these

Q.11

The number of ways in which a mixed double game can be arranged from amongst 9 married couples if no husband and wife play in the same game is

(A) 756

(B) 1512

(C) 3024

(D) none of these

Q.12

The number of permutations of letters of word HINDUSTAN such that neither the pattern 'HIN' nor 'DUS' nor 'TAN' appears are

(A) 12246

(B) 169194

(C) 166680

(D) 181434

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Q.13

The number of different words that can be formed from the letters of the work 'PENCIL' so that no two vowels are together is

(A) 120

(B) 260

(C) 240

(D) 480

Q.14

The set $S := \{1, 2, 3, \ldots\ldots, 12\}$ is to be partitioned into three sets A, B, C of equal size. Thus, $A \cup B \cup C = S, A \cap B = B \cap C = A \cap C = \phi$. The number of ways to partition S is

(A) $\frac{12!}{3!(4!)^3}$

(B) $\frac{12!}{3!(3!)^4}$

(C) $\frac{12!}{(4!)^3}$

(D) $\frac{12!}{(3!)^4}$

Q.15

If ${}^nC_{r-1} = 36, {}^nC_r = 84$ and ${}^nC_{r+1} = 126$ then the value of r is

(A) 1

(B) 2

(C) 3

(D) none of these

Q.16

The greatest possible number of points of intersection of 8 straight lines and 4 circles is

(A) 32

(B) 64

(C) 76

(D) 104

Q.17

There are 10 persons named $A, B, \ldots\ldots J$. We have the capacity to accommodate only 5. In how many ways can we arrange them in a line if A is must and G and H must not be included in the team of 5

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(A) ${}^8P_5$

(B) ${}^7P_5$

(C) ${}^7C_3(4!)$

(D) ${}^7C_3(5!)$

Q.18

The 7 men and 4 women are to seated at a round table. If all the women have to sit together, then the number of ways are

(A) 7!.4!

(B) $6!.3!$

(C) $10! - 7!.4!$

(D) none of these

Q.19

The number of ways of distributing 8 identical balls in 3 distinct boxes so that none of the boxes is empty is

(A) 5

(B) 21

(C) 3^8

(D) ${}^8\mathrm{C}_3$

Q.20

The number of non-negative integral solutions of $x + y + z = 5$ is same as

(i) number of ways of distributing 5 identical balls in 3 distinct boxes

(ii) number of selections of 5 objects from a lot containing 3 varieties of objects

(iii) number of selections of 3 objects from a lot containing 5 distinct objects

(A) (i) and (ii)

(B) (ii) and (iii)

(C) (i) and (iii)

(D) all of these

Q.21

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A student is allowed to select at most n books from a collection of $(2n+1)$ books. If the total number of ways in which they can select at least one book is 63, then the value of n is

Q.22

If the letters of the word SACHIN are arranged in all possible ways and these words are written out as in the dictionary, then the word SACHIN appears at serial number

Q.23

The number of non-negative integral solutions of the equation $x+y+3z=33$ is

Q.24

The sides AB, BC, CA of a triangle ABC, have 3,4 and 5 interior points respectively on them. The total number of triangles that can be constructed by using these points as vertices is

Q.25

If ${}^{12}P_r = {}^{11}P_6 + 6{}^{11}P_5$, then r is equal to

Q.26

The tens digit of $1! + 2! + 3! + \ldots\ldots + 49!$ is

Q.27

20 persons are invited for a party. The different number of ways in which they can be seated at a circular table with two particular person seated on either side of the host is K, find K/18!.

Q.28

The number of positive integral solutions of $abc = 30$ is

Q.29

The number of positive integers n such that 2^n divides $n!$ is

Q.30

In how many ways can you put 7 different letters into their respective envelopes such that exactly 3 go into the right envelope?

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Q.1 Answer:

$2^n - 2$

Solution:

Each object can be put either in box B_1 (say) or in box B_2 (say). So, there are two choices for each of the n objects. Therefore the number of choices for n distinct objects is $2 \times 2 \times \ldots . \times 2 = 2^n$ the second box being empty. Thus, there are $2^n - 2$ ways in which neither box is empty.

Q.2 Answer:

18

Solution:

Since we want 4 points for a parallelogram, two each on sets of parallel lines perpendicular to each other, the required number $= {}^4C_2 \times {}^3C_2 = 18$

Q.3 Answer:

${}^{21}C_7$

Solution:

Let x_i denotes the marks assigned to the 8 question.

Then, $x_1 + x_2 + x_3 + x_4 + x_5 + x_6 + x_7 + x_8 = 30$, where $x_i \geq 2, i = 1, 2, \ldots \ldots, 8.$

Let $y_i = x_i - 2, i = 1, 2, \ldots \ldots, 8.$

Then, $y_1 + y_2 + y_3 + \ldots \ldots + y_8 = 14$

The total number of solutions of this equation is ${}^{14+8-1}C_{8-1} = {}^{21}C_7$.

Q.4 Answer:

310

Solution:

(1) The number of four-digit numbers Strating with 5 is equal to $6^3 = 216$
(2) Starting with 44 and 55 is equal to $36 \times 2 = 72$
(3) Starting with 433,434 and 435 is equal to $6 \times 3 = 18$
(4) Remaining numbers are $4322, 4323, 4324, 4325$ is equal to 4 so total number are
$716 + 72 + 18 + 4 = 310$

Q.5 Answer:

80

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Solution:

Select 4 pairs in $^5C_4 = 5$ **ways**

Now select exactly one shoe from each of the pair selected in $\left(^2C_1\right)^4$ **ways**

$\Rightarrow$ **Required answers** $= 5 \times 16 = 80$

Q.6 Answer:

1240

Solution:

The number of ways of choosing first couple is $\left(^{15}C_1\right)\left(^{15}C_1\right) = 15^2$

The number of ways of choosing 2nd **couple is** $\left(^{14}C_1\right)\left(^{14}C_1\right) = 14^2$

Thus, the number of ways of choosing the couples is

$$15^2 + 14^2 + 13^2 + \ldots\ldots + 2^2 + 1^2$$
$$\frac{15\times(15+1)[2(15)+1]}{6} = 1240$$

Q.7 Answer:

5

Solution:

Number of groups having 4 boys and 1 girl $= \left(^4C_4\right)\left(^gC_1\right) = g$ **and**

number of groups having 3 boys and 2 girls $= \left(^4C_3\right)\left(^gC_2\right) = 2g(g-1)$

Thus, the number of dolls distributed $= g(1) + (2)[2g(g-1)] = 4g^2 - 3g$

Given the equation $4g^2 - 3g = 85 \Rightarrow g = 5$

Q.8 Answer:

882

Solution:

First we give one chocolate to each student after that remaining 42 chocolates can be distributed such that a student can get any number of chocolates can be understood by this equation $\mathrm{x} + \mathrm{y} + \mathrm{z} = 42$**, where** x**,** y **and** z **are non negative integers, so total number of solution is** $\begin{pmatrix} 44 \\ 2 \end{pmatrix}$**. Number of ways all three students getting an equal number of chocolates** $= 1$**. Number of ways exactly two of them are getting same number of chocolates** $= 3 \times 21 = 63$**.**

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Hence the number of ways when no two student receives same number of chocolates $= 44\mathrm{C}_2 - 1 - 3 \times 21 = 946 - 64 = 882.$

Q.9 Answer:

$6n - 3$

Solution:

first lets make a diamond necklace with the $2\mathrm{n}$ **diamonds**

Since all diamonds are identical there is only one possible way of doing this

Consider case of 3 different jewels of which 2 of them are always together

Number of ways to select 2 of the 3 jewels $= {}^3\mathrm{C}_2$

For 2 n diamonds we have $2n$ **vacant spaces where these two jewels can be placed**

As diamonds are identical wherever we place these two gems we will get same necklace. Hence there is only way

If one of $2\mathrm{n}$ **vacant spaces is filled with gem we have left with** $2\mathrm{n} - 1$ **vacant**

spaces

total number of possible necklaces $= {}^3\mathrm{C}_2 \times 2\mathrm{n} - 1$

$= \frac{3\times2}{2} \times (2\mathrm{n} - 1)$

$= 6\mathrm{n} - 3$

Q.10 Answer:

$16! - 1$

Solution:

$t_n = n(n!) = \{(\mathrm{n}+1) - 1\}\mathrm{n}! = (\mathrm{n}+1)! - \mathrm{n}!$ **which is now a telescopic sum.**

Q.11 Answer:

1512

Solution:

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Select 2 men from 9 men in ${}^{9}C_{2}$ ways. Then their wives are not to be selected therefore females will be selected from 7 in ${}^{7}C_{2}$ ways.

Now, 2 men and 2 women whence the mixed double can be played in 2 distinct ways.

$\Rightarrow$ **Total number of ways** ${}^{9}C_{2} \times {}^{7}C_{2} \times 2 = 1512$

Q.12 Answer:

169194

Solution:

Let A_1 be the property that HIN appears

Let A_2 be the property that DUS appears

Let A_3 be the property that TAN appears $n(A_1) = 7! = 5040$

$n(A_2) = \frac{7!}{2!} = 2520$

$n(A_3) = 7! = 5040$

$n(A_1 \cap A_2) = n(A_2 \cap A_3) = n(A_3 \cap A_1) = 5! = 120$

$n(A_1 \cap A_2 \cap A_3) = 3! = 6$

$n(A_1 \cup A_2 \cup A_3) = \sum n(A_i) - \sum_{i \neq j} n(A_i \cap A_j) + n(A_1 \cap A_2 \cap A_3)$

$= 12246$

Total number of ways to arrange the letters of the word HINDUSTAN $= \frac{9!}{2!} = 181440$

$\therefore$ Number of ways in which neither the pattern HIN or DUS and TAN appears

$= 181440 - 12246 = 169194$

Q.13 Answer:

480

Solution:

Using gap method, we first arrange the 4 consonants in $4!$ ways and the two vowels and be assigned gaps with arrangement in $\binom{5}{2}$ **$2!$ ways therefore total number of ways =**

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$4!^5P_2 = 48\mathrm{C}$

Q.14 Answer:

$\frac{12!}{(4!)^3}$

Solution:

12 different objects are to be divided into 3 groups of equal size, which are named as set A, B and C.

$\Rightarrow$ **Number of ways** $= \frac{(12)!}{(4!)^3 \cdot 3!} \times 3! = \frac{(12)!}{(4!)^3}$

Q.15 Answer:

3

Solution:

$\frac{^2\mathrm{C}_2}{^2\mathrm{C}_{\mathrm{r}-1}} = \frac{84}{36}$

$\rightarrow \frac{\frac{n!}{r'(n-r)!}}{\frac{n!}{(r-1)!(n-r+1)!}} = \frac{84}{36}$

$\rightarrow \frac{m-r+1}{r} = \frac{7}{3}$

$\rightarrow 3\mathrm{n} - 3\mathrm{r} + 3 = 7\mathrm{r}$

$\rightarrow 10\mathrm{r} = 3\mathrm{a} + 3\ \mathrm{m} \ldots\ldots\ldots\ldots\ldots + -1$

$\frac{^2C_{z+1}}{^2C_7} = \frac{126}{84}$

$$\frac{n!}{(r+1)!(n-r-1)!}$$

$$\rightarrow \frac{n!}{(r)t(n-r)!}\frac{3}{2}$$

$$\rightarrow \frac{n-r}{r+1} = \frac{3}{2}$$

$\rightarrow 5r - 2r - 3r + 3$

Dividing Equation 1 by Equation 2 ,

$\frac{10r}{5x} = \frac{3a-3}{2x-3}$

$\rightarrow 2(2\mathrm{n} - 3) = 3\mathrm{n} + 3$

$\rightarrow 4\mathrm{n} - 6 = 3\mathrm{n} + 3$

$\rightarrow \mathrm{n} = 6 + 3$

$\rightarrow \mathrm{n} = 9$

$10\mathrm{r} = 3\mathrm{n} + 3$

$\rightarrow 3(9) + 3 = 30$

$\rightarrow 10\mathrm{r} = 30$

$\rightarrow \mathrm{r} = 3$

$\therefore \mathrm{n} = 9, \mathrm{r} = 3$

Q.16 Answer:

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104

Solution:

8 straight lines meet at a point in $8C2$ ways $= 28$

Possible intersection points for lines and circles $= 8C1 \times 4C1 \times 2 = 64$

Intersection of two circles in two points $= 4C2 \times 2 = 104$

Therefore, the greatest possible points of intersection is 104

Q.17 Answer:

$^7C_3(5!)$

Solution:

Out of 10 persons, A is in and G and H are out of the team, so we have to select 4 more from 7 remaining.

This can be done in $^7C_4$ ways. These 5 persons can be arranged in a line in $5!$ ways.

Hence the number of possible arrangements is $^7C_4.5! = {}^7C_3(5!)$

Q.18 Answer:

7!.4!

Solution:

Considering the group of women as one entity, the circular permutation of 8 objects is $7!$ and women can arrange in $4!$ So (a) is the right choice.

Q.19 Answer:

21

Solution:

Number of ways is given by coefficient of x^8 in $\left(x^1 + x^2 + x^3 + \ldots.. + x^6\right)^3$ = coefficient of x^5 in $\left(x^0 + x^1 + x^2 + \ldots x^5\right)^3$

= coefficient of x^5 in $(1-x)^{-3}\left(1-x^6\right)^3$

= coefficient of x^5 in $(1-x)^{-3}$

=$^7C_2 = 21$

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Q.20 Answer:

(i) and (ii)

Solution:

Number of non negative integral solutions of $x+y+z=5 \begin{pmatrix} 5+3-1 \\ 3-1 \end{pmatrix} = \begin{pmatrix} 7 \\ 2 \end{pmatrix} \neq \begin{pmatrix} 5 \\ 3 \end{pmatrix}$

Q.21 Answer:

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Solution:

The problem lead to ${}^{2n+1}C_1 + {}^{2n+1}C_2 + {}^{2n+1}C_3 + \ldots\ldots + {}^{2n+1}C_n = 63$

From the usual identity

$${}^{2n+1}C_0 + {}^{2n+1}C_1 + {}^{2n+1}C_2 + \ldots.. + {}^{2n+1}C_n + {}^{2n+1}C_{n+1} + {}^{2n+1}C_{n+2} + \ldots.. + {}^{2n+1}C_{2n+1} = 2^{2n+1}$$

$$2\left[{}^{2n+1}C_0 + {}^{2n+1}C_1 + {}^{2n+1}C_2 + \ldots\ldots + {}^{2n+1}C_n\right] = 2^{2n+1} \quad \therefore \quad {}^{2n+1}C_r = {}^{2n+1}C_{2n+1-r}$$

$$\Rightarrow \quad 2[1+63] = 2^{2n+1} = 2^7 = 2^{2n+1} \Rightarrow n = 3$$

Q.22 Answer:

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Solution:

Words starting with $\mathrm{A}, \mathrm{C}, \mathrm{H}, \mathrm{I}, \mathrm{N}$ **are each equals to** $5!$

$\therefore$ **Total words** $= 5 \times 5! = 600$

The first word starting with S **is SACHIN**

$\therefore$ **SACHIN appears in dictionary at serial number** $601.$

Q.23 Answer:

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Solution:

$x+y+3z = 33 \Rightarrow x+y = 33-3z$

Let $z = k.$

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Then, $x+y=33-3k$.

The number of non-negative integral solutions of $x+y=33-3z$ is

$^{33-3k+2-1}C_{2-1}={}^{34-3k}C_1=(34-3k)$

But $0\le 33-3k\le 33\Rightarrow 0\le k\le 11$

Hence, total number of solutions =$\sum_{\mathrm{n}}=\sum_{k=0}^{11}(34-3k)=\frac{12}{2}(34+1)=210$

Q.24 Answer:

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Solution:

We have in all 12 points.

Since 3 points form a triangle.

$\therefore$ **the total number of triangles, including the triangles formed by collinear points on**

AB, BC **and** CA**, is** $^{12}C_3=\frac{12\times 11\times 10}{1\times 2\times 3}=220$

But this includes the number of triangles formed by 3 points on AB **i.e.** $^3C_3=1$ **and**

for 4 points on $BC={}^4C_3=4$ **and 5 points on** BC **i.e.** $^5C_3=\frac{5\times 4}{1\times 2}=10$

$\therefore$ **Required number of triangles** $=220-1-4-10=205$

Q.25 Answer:

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Solution:

$r\cdot {}^{n-1}P_{r-1}+{}^{n-1}P_r={}^nP_r$

$$\begin{aligned}\therefore\quad & 6\cdot {}^{11}P_5+{}^{11}P_6={}^{12}P_6\\ \Rightarrow\quad & {}^{12}P_r={}^{12}P_6\quad \left(6\cdot {}^{11}P_5+{}^{11}P_6={}^{12}P_r \text{ given }\right)\\ \Rightarrow\quad & r=6\end{aligned}$$

Q.26 Answer:

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Solution:

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We have $1! + 2! + 3! + 4! = 33$.

Also $5! = 120, 6! = 720, 7! = 5040, 8! = 40320$ and $9! = 326880$.

Thus, the tens digit of $1! + 2! + \ldots + 9!$ is 1 .

Also, note that $n!$ is divisible by 100 for all $n \geq 10$,

so that ten digits of $10! + 11! + \ldots . + 49!$ is zero.

Therefore, tens digit of $1! + 2! + \ldots \ldots + 49!$ is 1.

Q.27 Answer:

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Solution:

Total number of persons in party $= 21$ As two persons A and B are to be seated besides the host H, number of arrangements are AHB and BHA. Taking $\{A, H, B\}$ as a single element. no of person $\mathrm{s} = 19$

$\therefore$ no of arrangements $= (\mathrm{n} - 1)! = 18!$

$\therefore$ Total no of ways $= 18! \times 2$ ways of arrangement besides host $= 18! \times 2$

Q.28 Answer:

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Solution:

We have, $30 = 2 \times 3 \times 5$. So 2 can be assigned to either a or b or c i.e. 2 can be assigned in 3 ways. Similarly, each of 3 and 5 can be assigned in 3 ways. Thus the no. of solutions are $3 \times 3 \times 3 = 27$

Q.29 Answer:

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Solution:

The exponent of 2 in $n!$ is given by

$E = \left[\frac{n}{2}\right] + \left[\frac{n}{2^2}\right] + \left[\frac{n}{2^3}\right] + \ldots$

where $[x]$ denotes greatest integer $\leq x$, As $[x] \leq x \forall x$ $\left[\frac{n}{2^m}\right] = 0$ after finite number of terms. Thus we get

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$E < \frac{n}{2} + \frac{n}{2^2} + \frac{n}{2^3} + \ldots = \frac{n/2}{1-1/2} = n$

Thus, there is no positive integer for which 2^n divides $n!$

Q.30 Answer:

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Solution:

No of ways in which the 3 correct envelopes can be selected $= \begin{pmatrix} 7 \\ 3 \end{pmatrix} = 35$

Derangement of the remaining 4 envelopes & letters $= 9$ **(derangement value for 4 is 9)**

Total No of ways of arrangement $= 9 \times 35 = 315.$

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